OAK RIDGE National Laboratory

AI-Agent Driven Survey and Validated Literature Review for Discovering Solid Breeder Materials

End-to-End Traceability for Automated Literature Generation

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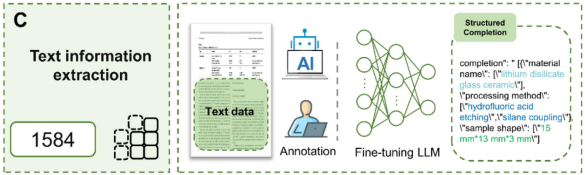
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U.S. DEPARTMENT OF ENERGY

AI-Agent Driven Literature Reviews with Full Claim Traceability



End-to-End Generation



OCR with bounding-box capture

- **Manual reviews**: Screening hundreds of papers can take weeks to months.
- **Commercial platforms**: Subscription fees, token limits, and usage caps restrict scalability.
- **LLM-generated reviews**: Often lack transparent source attribution and traceability.

Our contribution: a LLM pipeline with per-claim traceability.

Solid Breeders are one the Bottleneck for Fusion Energy

Current Approaches

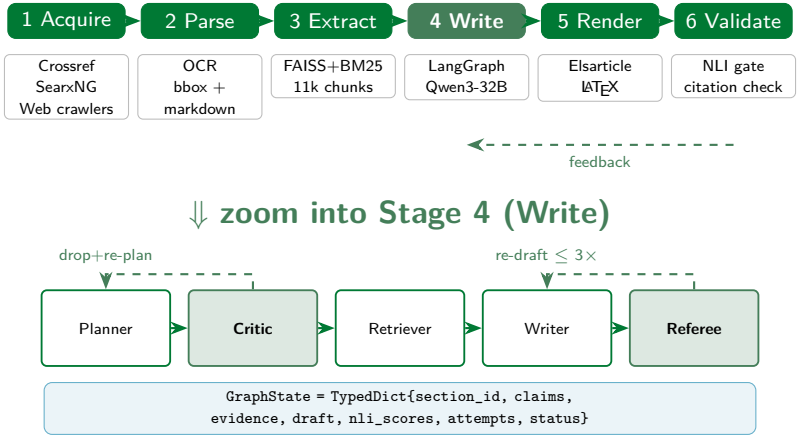
- Analyze papers and databases individually
- Limited claim-level verification
- Static LLM prompts cannot explore research questions in depth
- Focus on prediction, not review generation

Our Approach

- Integrate multimodal information (incl. ontologies)
- Verifies every sentence
- Produces traceable review manuscripts automatically
- Graph-based approach enables deeper exploration of research questions

References: Zhou *et al.*, *Int. Dent. J.* 2026 doi:10.1016/j.identj.2025.07.013 ○ Tasdemir, *J. Alloys Metall. Syst.* 2026 doi:10.1016/j.jalmes.2025.100192 ○ Tang *et al.* (MatterChat), *Nat. Mach. Intell.* 2026 doi:10.1038/s42256-025-01065-z ○ Zhang *et al.*, *Adv. Funct. Mater.* 2026 doi:10.1002/adfm.202508472.

Six Stages End-to-End | Writing Stage Expands Into a 5-Agent LangGrap With Shared State

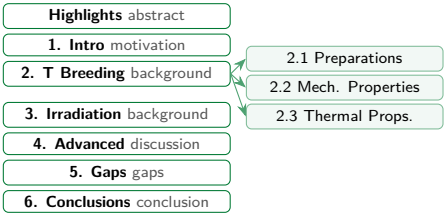


Why a graph: The graph breaks broad scientific questions into focused sub-questions, while a shared state records sources and decisions for per-claim traceability.

Ontologies Tell Agents What and How to Write

Structure ontology: What to write

18 sections extracted from the reference paper:



Each node → {role, purpose, expected_evidence, research_questions}.

Style ontology: How to write

```
hedging:  medium

citation:  numeric
```

Indexed Chunk: What to retrieve

```
Material:Li2TiO3
Property:tritium.T50
Condition:temp_K=673

bbox=[72,310,540,442]
```

How both ontologies are produced:

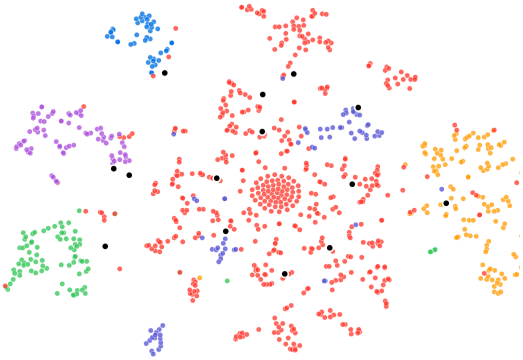


Domain expert edits both before any agent runs.

Critic and Referee Coverage are Structured Before Writing

Corpus embedding

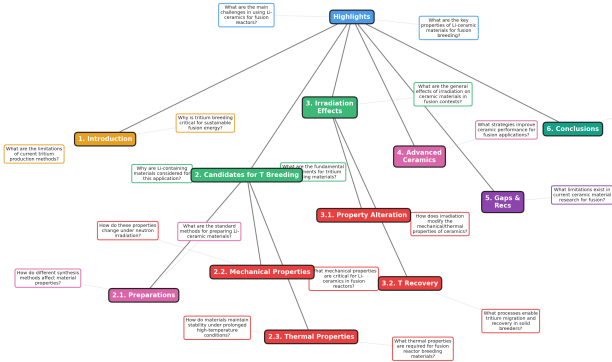
984 chunks · 14 papers · TF-IDF + t-SNE, k-means (k=6)



- hidrojen / hte / hydrogen
- systems / injection / plasma
- tritium / inventory / reserve
- fusion / left / fuel
- fusion / nucl / org
- fueling / source / plasma

dots = research direction.

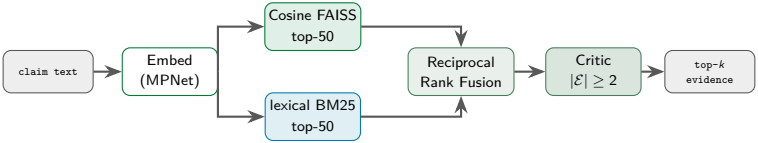
Research-direction outline



18-section outline extracted from the reference paper.

Per-Claim Paper Selection: Hybrid Lexical and Semantic Retrieval: Evidence Search

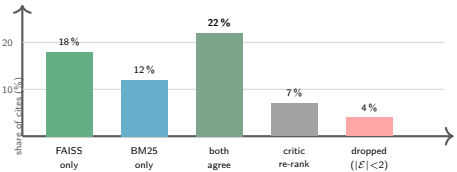
Critic (pre-write): $|\mathcal{E}_i| \geq 2$ with distinct sources, else drop. **Referee** (post-write): $P_{\text{ent}} \geq 0.55 \wedge |\hat{y} - y|/|y| \leq 0.10$, else re-draft. **Composite:** $Q = 0.3\,S_g + 0.3\,V_{\text{cite}} + 0.2\,S_f + 0.2\,S_{\text{sty}}$.



Entailment score $P_{\text{ent}}(p_j \mid s_j)$:
probability that source chunk s_j logically supports pipeline sentence p_j (DeBERTa-v3 NLI).

Accept iff:
- $P_{\text{ent}} \geq 0.55$
- numeric error $|\hat{y} - y|/|y| \leq 0.10$
- citation overlap $C_j \geq 0.5$

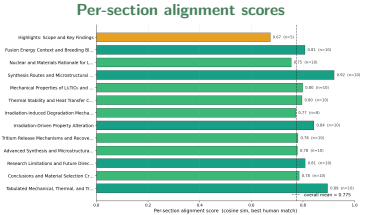
values logged to `retrieval_audit.json`



Per-Claim Traceability in Action: Accepted Claims are Incorporated. Failed Claims Are Dropped (Referee)

Pipeline claim (with surrounding context)	Human expert (BNT review, with context)	E	Source
. . . upon neutron capture, Li_2TiO_3 T release peaks ∼673 K, *enabling efficient breeding.*	*Out-of-pile experiments report* the tritium release peak near 400 °C *for* Li_2TiO_3 *pebbles.*	**0.87**	[42]p.3
Microstructure refines: grain $<1\ \mu$m after 0.1 dpa, *raising swelling resistance.*	*Neutron irradiation campaigns show* sub-micron grain refinement above 0.05 dpa, *consistent with TEM observations.*	**0.79**	[17]p.7
Thermal conductivity κ drops 35 % at 0.3 dpa, *limiting heat extraction.*	*Post-irradiation examination reports a* 30–40 % reduction in thermal conductivity *at comparable dose levels.*	**0.82**	[23]p.12
Claim asserts Li_2ZrO_3 superior above 900 °C *– no supporting evidence found.*	*(BNT review discusses* Li_2ZrO_3 *stability but reports* no comparative performance above 900 °C.*)*	**0.41**	**drop**

Green/black: matched claim text; grey italic: surrounding context; **red**: entailment score E; last row dropped ($E < 0.55$).



Score	Value
Mean alignment	**0.77**
Pairs evaluated	167
Accepted (≥ 0.55)	138
Dropped	29
Entailment E	0.81

Ground truth: BNT expert review.

Entailment E: probability that the source chunk logically supports the pipeline sentence (DeBERTa-v3 NLI). Accept iff $E \geq 0.55$, numeric error ≤ 0.10, citation overlap ≥ 0.5.

Side-by-Side Comparison: Human and Pipeline Manuscripts Agree on the Same Claims

Human-expert manuscript

Expert review, p. 1

this element in nature. Production of tritium is achieved by neutron irradiation of lithium (Li) containing materials. Prominent tritium producing materials, known as breeders, are classified as either solids or liquids. The review here focuses on solid materials such as Li_2TiO_3, Li_2ZrO_3, Li_4SiO_4, and advanced

Performance of irradiation campaigns using neutron, ion beam, and other sources are explored as well as exploration of the literature for advanced ceramics such as biphasic or 3D printed materials. Utilization of

influence on grain structure, density, porosity, and phase stability. Examination of literature offers insight into the mechanical and thermal properties of solid breeders and the irradiation induced alterations.

Pipeline-generated manuscript

paper.pdf, p. 10–11

The production of tritium via neutron irradiation of lithium-containing solid breeders, such as Li_2TiO_3 and Li_2ZrO_3, is a critical component of fusion energy systems. It is essential to understand the chemical, mechanical,

alterations.

The performance of irradiation campaigns using neutron, ion beam, and other sources provides insight into how these materials respond to flux. It appears that the introduction of defects during production methods-such

diation conditions [3]. Neutron irradiation damage in solid breeders is an important issue that necessitates a comprehensive examination of literature regarding mechanical and thermal properties alongside irradiation-induced

Metric	Score	Metric	Score	Metric	Score
Claim recall	0.78	Faithfulness (NLI)	0.94	JSD coverage	0.12
Claim precision	0.71	Citation grounding	1.00	NMAE numerics	0.09
				Composite Q	**0.79**

The Full Stack Runs Locally on One GPU - One Click Deployment



Docker containers orchestrate the stack.
Ollama serves local LLM inference.

Zotero (ref+PDF)

SearXNG (search)

Crawl4AI

↓

MinerU (GPU OCR)

↓

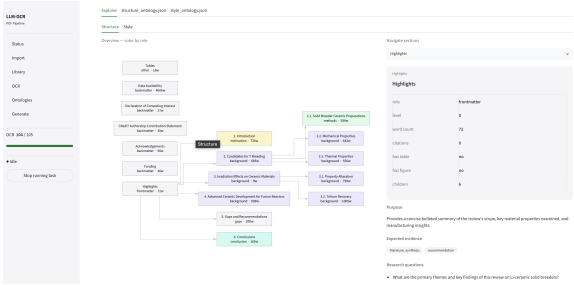
Ollama (Qwen3-32B)

↓

Pipeline (LangGraph)

↓

Streamlit UI :7860



Browser interface: Ontology editor, pipeline status, LaTeX manuscript.

Public images

- hub.docker.com/r/jtupayac/llm-ocr-pipeline
- hub.docker.com/r/jtupayac/llm-ocr-ui

Questions?

Backup: Validation Metrics

Formal definitions

Metric	Formula
Claim recall	$R = \dfrac{\lvert\mathcal{C}_h \cap \mathcal{C}_p\rvert}{\lvert\mathcal{C}_h\rvert}$
Claim precision	$P = \dfrac{\lvert\mathcal{C}_h \cap \mathcal{C}_p\rvert}{\lvert\mathcal{C}_p\rvert}$
Faithfulness (NLI)	$F = \dfrac{1}{n}\sum_j P_{\text{ent}}(p_j \mid s_j)$
Citation grounding	$G = \dfrac{\lvert\{j : \text{bbox}_j \text{ verified}\}\rvert}{\lvert\mathcal{C}_p\rvert}$
Topic divergence	$\text{JSD}(H \,\Vert\, P)$ over LDA topic mixtures
Numeric error	$\text{NMAE} = \dfrac{1}{n}\sum_j \dfrac{\lvert\hat{y}_j - y_j\rvert}{\lvert y_j\rvert}$
Composite quality	$Q = 0.3\,S_g + 0.3\,V_{\text{cite}} + 0.2\,S_f + 0.2\,S_{\text{sty}}$

In plain language

Metric	In five words
Claim recall	human facts we recover
Claim precision	our facts matching the human
Faithfulness	sentences supported by source chunk
Citation grounding	claims with verified source bbox
Topic divergence	divergence of topic emphasis
Numeric error	mean normalized numeric deviation
Composite Q	weighted average of all metrics

Notation. $\mathcal{C}_h$, $\mathcal{C}_p$: human and pipeline claim sets. p_j, s_j: pipeline sentence and source chunk. P_{ent}: DeBERTa-v3-MNLI entailment probability. Accept iff

Backup: Glossary of Acronyms, Models and Domain Terms Used Throughout the Talk

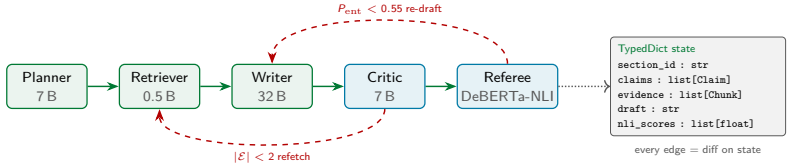
Methods and models

Term	Meaning
TF-IDF	Term Frequency $\times$ Inverse Document Frequency.
SVD	Singular Value Decomposition (TF-IDF $\rightarrow$ 50 dims).
t-SNE	t-Distributed Stochastic Neighbor Embedding (2D vis).
LDA	Latent Dirichlet Allocation; topic-mixture model.
BM25	Okapi BM25 lexical relevance score.
FAISS	Facebook AI Similarity Search; dense vector index.
MPNet	Masked-and-Permuted sentence-embedding transformer.
RRF	Reciprocal Rank Fusion of multiple rankings.
NLI	Natural Language Inference; entailment classifier.
LangGraph	Typed-state agent-graph framework on LangChain.
MinerU	GPU PDF parser yielding markdown + bounding boxes.

Metrics and domain terms

Term	Meaning
JSD	Jensen–Shannon Divergence (0 = identical).
NMAE	Normalised Mean Absolute Error on numerics.
Q	Composite quality $0.3\,S_g{+}0.3\,V_{\text{cite}}{+}0.2\,S_f{+}0.2\,S_{\text{key}}$.
bbox	Bounding box (page, x_0, y_0, x_1, y_1) of a claim.
RQ	Research Question on a Structure-ontology node.
DT fusion	Deuterium–Tritium fusion: $^2\text{H}{+}^3\text{H} \rightarrow {}^4\text{He}{+}n$.
TBR	Tritium Breeding Ratio ($\geq$ 1.05 required).
dpa	Displacements per atom; neutron damage dose.
MHD	Magneto-Hydro-Dynamic pressure drop.
PbLi, FLiBe	Liquid breeder candidates (PbLi eutectic, LiF–BeF_2).
ITER, DEMO	ITER reactor + its demonstration successor.

Backup: Why a Graph Instead of a Single LLM Call?



Monolithic LLM call: prompt → paragraph × no typed state × full re-run on any failure × no per-claim audit

Backup: Validation Protocol: Every Sentence Is Scored Against a Human Expert

Alignment protocol

1. Split human & pipeline drafts into sentences.
2. For each p_j find nearest $h_{\pi(j)}$ in MPNet space.
3. Score the pair on three axes:

- **Entailment** E_j — DeBERTa-v3 NLI
- **Numeric error** $N_j = |\hat{y}_j - y_j| / |y_j|$
- **Citation overlap** C_j

Accept iff $E_j \geq 0.55 \wedge N_j \leq 0.10 \wedge C_j \geq 0.5$.

What each metric means

- **Recall** = % of human claims we found.
- **Precision** = % of our claims that match human.
- **Faithfulness** = NLI agreement to source chunks.
- **Citation grounding** = % of cited values with a verified bbox.
- **JSD** = topic-distribution divergence (0 = identical emphasis).
- **NMAE** = normalized numeric error.

Ground truth: 30-page expert review on Li_2TiO_3 breeders. 614 H | 587 P | 478 aligned pairs.